
EE/CprE/SE 491 WEEKLY REPORT 7

Apr 8, 2025 - Apr 15, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates

Daniel Vergara Pinilla

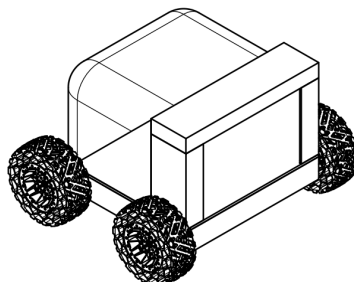
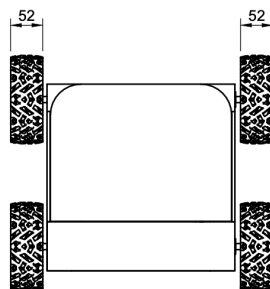
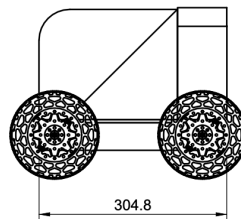
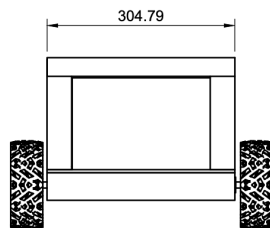
Mitchell D Owen

Om Shukla

Sudipta Halder

○ Weekly Summary

This week we made a quick design change due to our goal to make the robot under five hundred dollars. We will be shrinking the size of the robot to 1 foot by 1 foot. This way we can achieve a high quality of components for a smaller size while being cheaper. This will make it easier to build. Also we have decided to make an IOS and web app for user interface.



○ **Past week accomplishments**

- Sudipta Halder: Completed adding motors to BoM. created the first iteration of the fusion cad model of our robot
- Brodie Bates: Found linear actuators, started researching microcontrollers.
- Daniel Pinilla: Worked on the IOS application for the end user interaction
- Om Shukla: Researching geofencing and learning about front-end website development.
- Mitchell Owen: Researching microcontrollers and making a survey for user testing.

○ **Pending issues**

- Mitchell Owen: Falling behind assembling the robot. We need to start testing soon.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Research Linear actuators, batteries, and sensors	4	39
Om Shukla	Researching geofencing and learning about front-end website development.	4	31
Sudipta Halder	Worked on finding part for BoM, made first iteration of cad models. Created a top down level of the roomba.	6	45
Daniel Mauricio Vergara	Worked on the UI and UX of the IOS app for User interaction	7	39
Mitchell Owen	Researching microcontrollers. And make survey for user testing	4	34

○ **Plans for the upcoming week**

- Sudipta Halder: simulated some of the motors and parts we are going to use in Webots. Complete Bom. Create a cheap low powered linear actuator.
- Brodie Bates: Create basic embedded program to control motors and linear actuators and help decide final specs on motors, sensors, and actuators.
- Om Shukla: Learn more about front-end development, and set up a webpage for user login with working login credentials from the server.
- Mitchell Owen: Find all the components for the robot and start assembly.
- Daniel Mauricio Vergara: Finish web socket communication and API with Google Maps to provide specific locations of the Robot all the time

○ **Summary of weekly advisor meeting**

No meeting advisor meeting from time period **Apr 8, 2025** - **Apr 15, 2025**